

Anomaly of Dielectric Constant of Ice

*Camila Raupp da Luz, Marcia Barbosa, Marco A. Habitzreuter, Eduardo O. Rizzatti
Universidade Federal do Rio Grande do Sul (UFRGS)

Even though the water constitutes $\frac{2}{3}$ of Earth's surface and is vital for life, it exhibits approximately 70 anomalous behaviors to be studied and understood. Many of these properties can be attributed to its hydrogen bonding and angular structure. At low temperatures, the quantum properties of hydrogen bonds become relevant to describe anomalies, as the case of the dielectric constant. The dielectric constant of water consists of a real part and an imaginary part. An experiment with ice shows that, unlike other materials, the imaginary part of the dielectric constant does not exhibit a monotonous decrease as temperature decreases as the real part does. Instead, it shows a minimum around 20 K followed by an increase as the temperature decreases. One possible explanation for this anomaly is the occurrence of proton collective tunneling within the hydrogen bonds of hexamers, which weakens these bonds. This phenomenon explains the reason for this behavior only in the imaginary part of the dielectric constant because it does not change the dipole moment of the hexamers. The present work aims to use molecular dynamics simulations to understand the anomaly of dielectric constant of ice, combining path integral molecular dynamics (PIMD) and adaptive resolution techniques. However, even classical models used to simulate ice cannot reproduce the correct value of dielectric constant, which this work is seeking to solve first. Also, in parallel to this, a simple lattice model of squared ice is proposed to analyze if collective tunneling can affect the dielectric constant.